

ShopSmart Retail Consulting Case

Individual Assessment

Scenario

You are a Business Consultant at **InsightAI Consulting**.

Your client, **ShopSmart Retail**, operates stores across multiple regions. Although revenue appears healthy, management is concerned that profits are not growing as expected. They suspect there may be operational issues related to inventory, labour, suppliers, or regional performance.

You have been given two datasets:

- **sales_transactions.csv**
- **store_info.csv**

The data contains quality issues and cannot be trusted without investigation.

Your task is to clean the data, analyze business performance, and provide recommendations to management.

Time Limit

120 Minutes

Deliverables

Submit to your github link and upload that in the classroom.

1. **clean_sales_analysis.csv**
2. **analysis.ipynb** (or analysis.py)
3. **Consulting Summary** (This can be in your analysis or a separate file)

Part A – Data Cleaning (25%)

Investigate and clean the datasets.

Possible issues include:

- Duplicate records
- Missing values
- Invalid quantities
- Inconsistent IDs
- Mixed date formats
- Mixed currency formats
- Inconsistent category and supplier names

Required Checkpoints

Report the following:

1. How many duplicate transactions were removed?
 2. How many invalid quantity records were removed?
 3. How many records remain after cleaning?
 4. What is the total revenue?
 5. What is the total gross profit?
-

Part B – Data Analysis & Visualization (25%)

Create at least **three visualizations**.

Suggested topics:

- Revenue by product category
- Profit by category
- Profit by region
- Stockout rate by region
- Labour efficiency by region
- Supplier performance

All visualizations should support your conclusions.